

NCATS OLIGOTOX OPEN DATA CHALLENGE · PHASE 2

# OligoTox-Kidney

A curated, per-measurement **kidney-toxicity** dataset for oligonucleotide therapeutics

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65 oligos · 111 measurements · 35 target genes · 100% kidney-specific

Phase 2 · Data Generation · prepared for biochemistry-expert review · June 2026

THE IDEA IN ONE SENTENCE

**Oligonucleotide drugs can injure the kidney in a way the standard safety test misses — so we built a dataset that captures the right signal, one experiment per row, every value traceable to its source.**

# Why this matters

**Oligonucleotides are a booming drug class** — ~19 already approved, dozens in trials, treating diseases nothing else could.

**But the kidney is a common casualty.** The kidney filters and concentrates these drugs, so it is exposed to high levels — and several oligos have caused proteinuria, kidney injury, even kidney failure.

**The catch:** the damage is often *invisible* to routine cell-survival screening (next slides).

## The gap Phase 2 targets

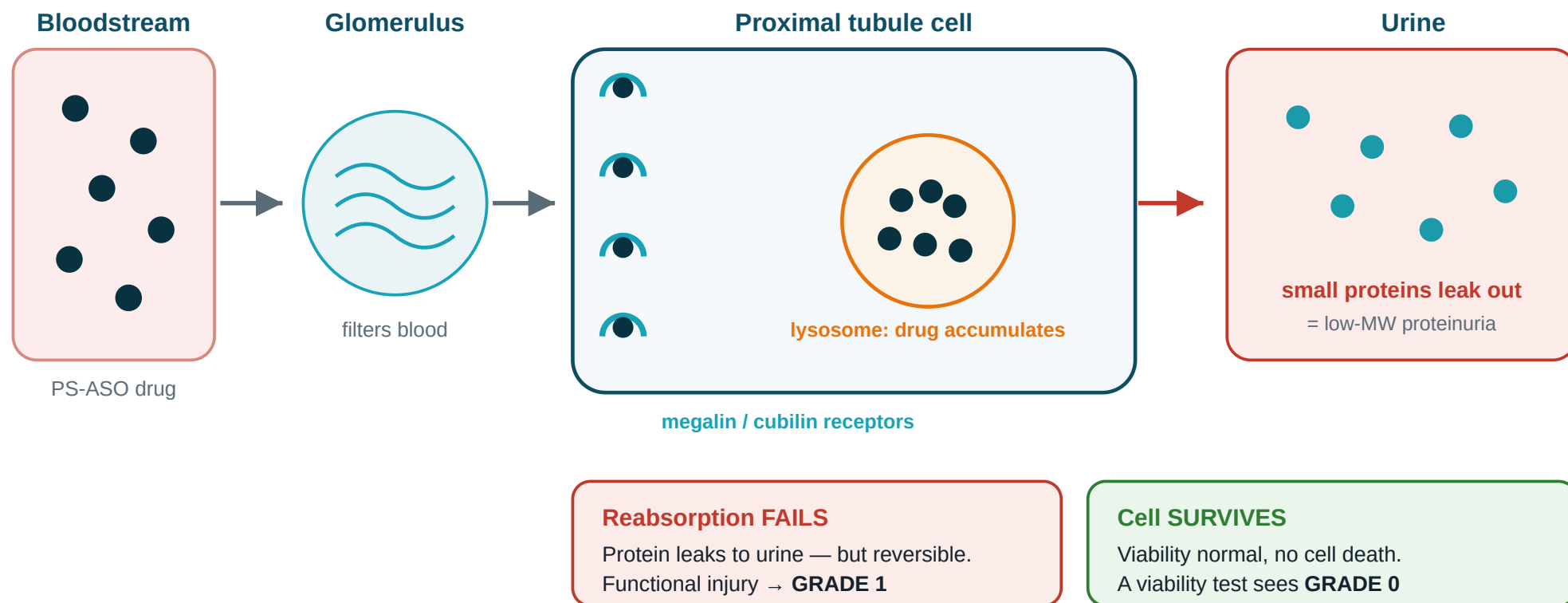
Most public oligo-tox data is **hepatic**; kidney-specific data is thin. No open dataset pairs each oligo's **design** (sequence + chemistry) with a **graded kidney outcome** — exactly what an **in-silico model** needs to predict risk early. That is what we assembled.

PART 1

# The science: a quiet kind of kidney injury

Why a cell-survival test gives these drugs a false "all-clear"

# How these drugs injure the kidney



Phosphorothioate ASOs are filtered, then vacuumed into proximal-tubule cells by the megalin/cubilin receptors and stored in lysosomes. Reabsorption of small proteins fails → they leak into urine (proteinuria) — yet the cell stays alive.

# The trap: one drug, two tests, opposite answers

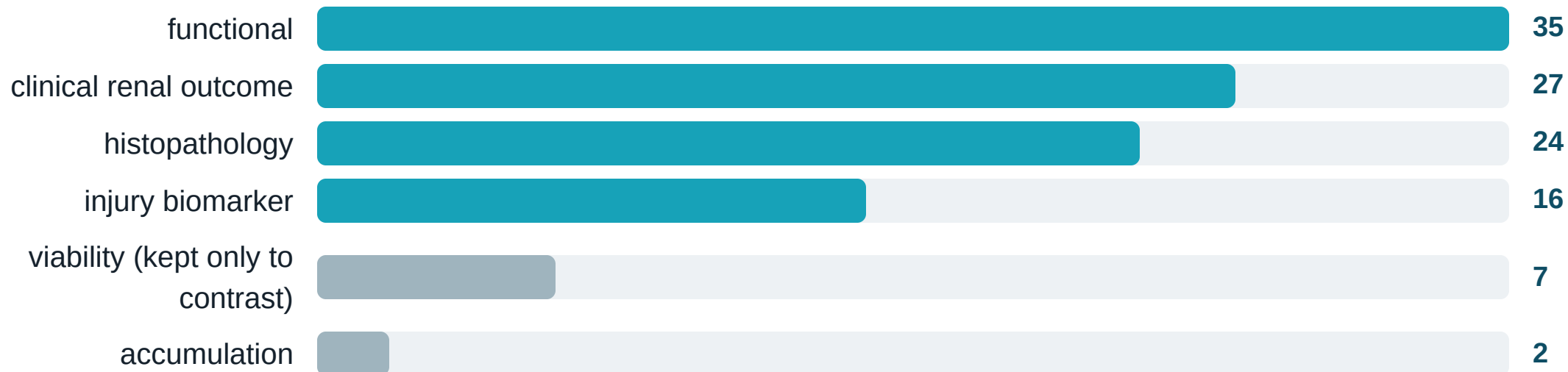
Same drug. Two tests. Opposite conclusions.



A viability/MTT screen asks only "**are the cells alive?**" — and these drugs don't kill cells. So routine screening calls them **clean**. Our schema is built to capture the **functional** signal that test misses.

## So we record the *right* readouts

We deliberately weight the data toward **function and injury markers**, not cell survival:



“Readout” = one measured thing. Biomarkers = chemicals (KIM-1, NGAL, clusterin, cystatin C) that rise when the kidney is stressed. Functional + biomarker rows (51) dwarf viability rows (7) — by design.

PART 2

# What we built

The dataset, its structure, and how toxicity is scored



## The dataset at a glance

**65**

unique oligos  
(7 drug families)

**111**

graded measurements  
(target was  $\geq 100$ )

**100%**

kidney-specific  
(no padding)

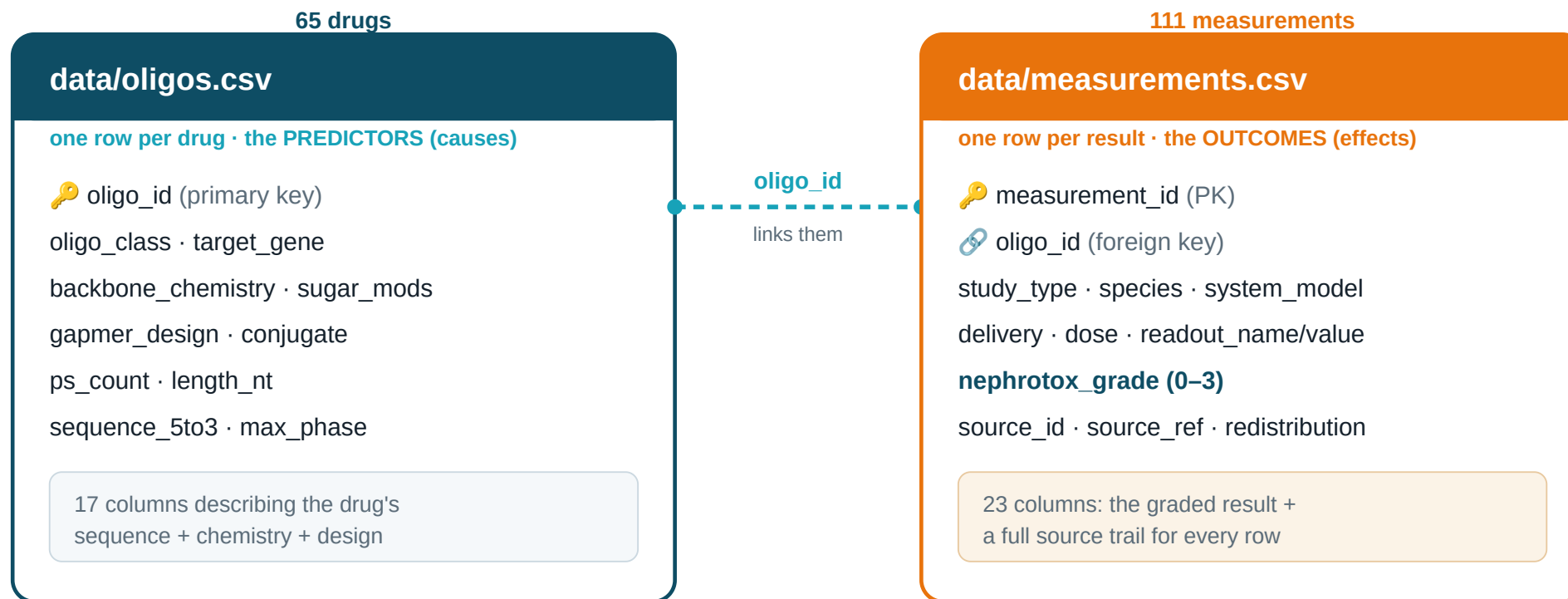
**16**

distinct sources  
fully cited

Spans every oligo modality, all three study types (dish / animal / human), and the full severity range from safe to severe — including deliberate negative controls.

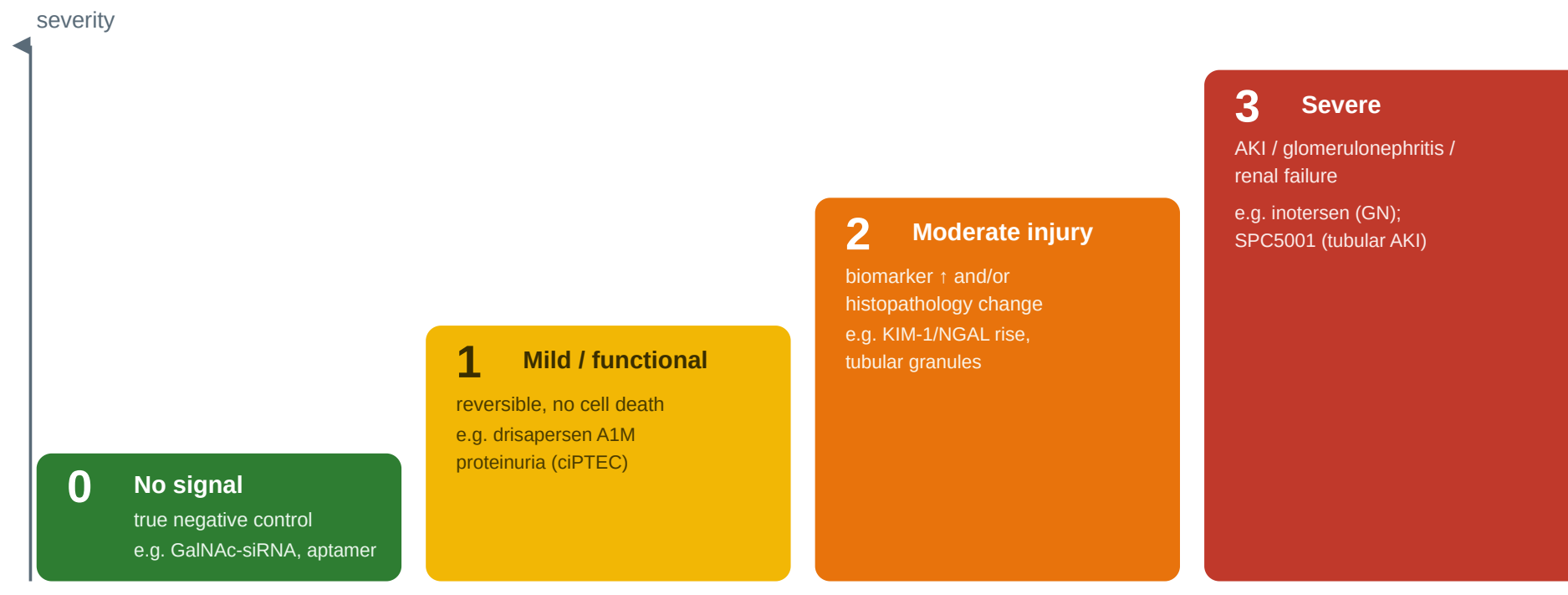
**35** target genes · **33/65** sequences filled (rest honestly marked “TBD”, never guessed) · every row carries its source.

# How the data is organised — two linked tables



One table describes each **drug** (the suspected causes); the other lists each **result** (the effects). They share an ID (**oligo\_id**) so any result links back to its drug — like two tabs in a spreadsheet joined by a key.

# The scoring system: a 0–3 severity grade



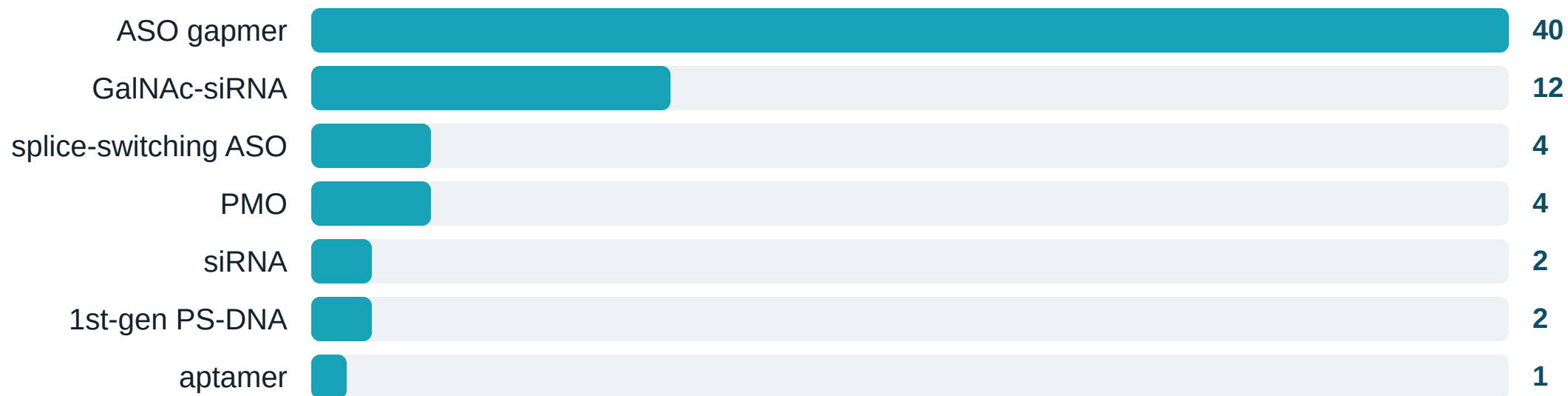
Each result gets one ordinal grade by a written rubric (in `schema.md`). **All grades are currently flagged “provisional”** — confirming them is the sign-off we’re asking a biochemistry expert for.

## Grades span the full range — including safe controls



A healthy spread across all four levels — a usable dataset needs examples of **every** severity, not just the dramatic ones. The 27 grade-0 rows are real negative controls (GalNAc-siRNA, intrathecal ASO, aptamer).

## Coverage: every oligonucleotide family



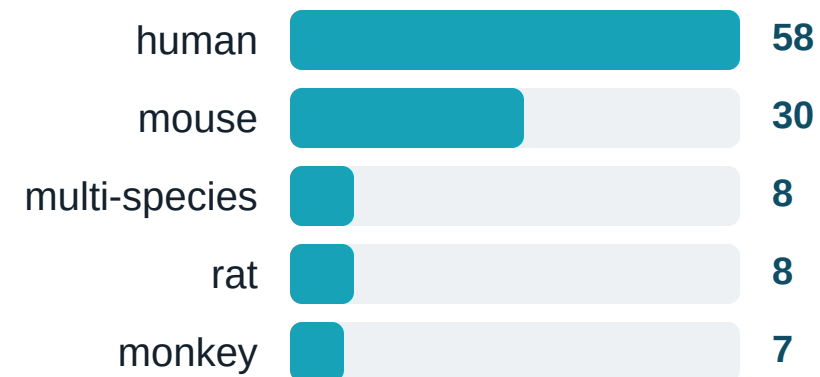
**Gapmer** = central DNA “gap” + chemically-modified wings (cuts target RNA). **siRNA** = double-stranded silencer; **GalNAc** = sugar tag aiming it at the liver. **PMO** = neutral-backbone splice fixer (the muscular-dystrophy drugs). **Aptamer** = an oligo folded to grab a protein, like an antibody.

## Coverage: dish → animal → human (the translation axis)

### STUDY TYPE



### SPECIES



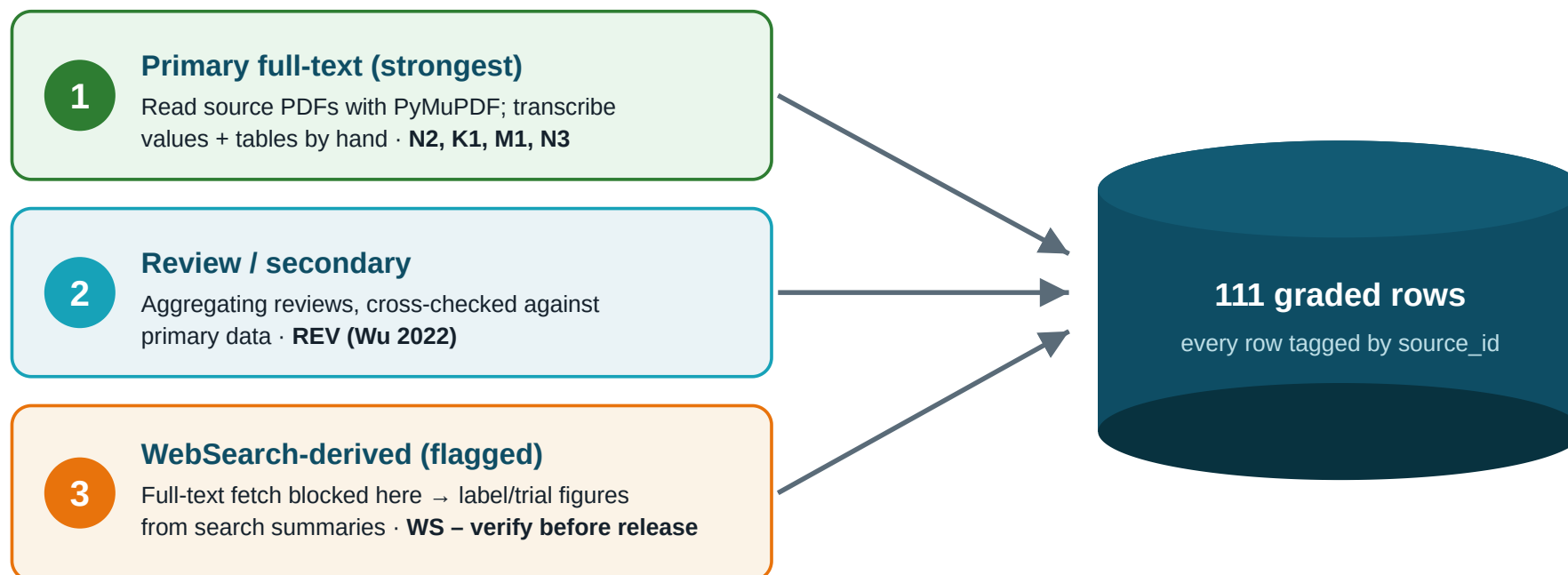
Having dish, animal, and human rows side by side — measured the same way — lets the **animal-to-human translation gap** be studied directly (Finding 3).

PART 3

# How we built it

Methodology, the integrity rule, and the sources researched

# Three ways we gathered data — each tagged per row



Every row records *how* it was obtained, so its reliability is visible. Web-search rows are flagged for re-checking because this secure environment blocks downloading full papers.



# The rule that makes the dataset trustworthy

## Strict no-fabrication policy

Drug **sequences** and toxicity **values** are never invented or recalled from memory. A value goes in **only** if an explicit, citable, redistribution-permitted source states it. Drugs with no published kidney data were **omitted, not padded**.

### WHY SEQUENCES ARE 33/65, NOT 65/65

The 32 blanks are **real gaps**, honestly marked “TBD”. We could have made the table look complete by guessing — but a falsely-complete table can’t be trusted at all.

### WHY THIS MATTERS TO A REVIEWER

If the easy fields were fudged, none of the hard toxicity values could be believed. Honest blanks are the **credibility** of the whole dataset.

# Sources researched

## STRICT-KIDNEY PRIMARY (READ IN FULL)

- **Janssen 2019** · drisapersen, human kidney-cell injury (*N2 · 10 rows*)
- **US 11,105,794 B2** · patent tox panel (*N3 · 21 rows*)
- **Moisan 2017** · human tubule-cell panel (*M1 · 11 rows*)
- **Sandelius 2020** · urine injury-biomarkers (*K1 · 9 rows*)
- **van Poelgeest 2013** · SPC5001 first-in-human AKI (*A3*)
- **Arch Toxicol 2021** · SPC5001 kidney-on-chip (*A4 · 5 rows*)

+ **36 “WS” rows** from FDA/EMA labels & pivotal trials: patisiran, vutrisiran, lumasiran, nedosiran, eplontersen, tofersen, bepirovirsen, olpasiran, the DMD PMOs, pegaptanib, fitusiran, zilebesiran — plus Crooke 2018 (human) & Yu 2012 (monkey) for the translation gap.

## ANCHORS · REVIEW · PATENTS

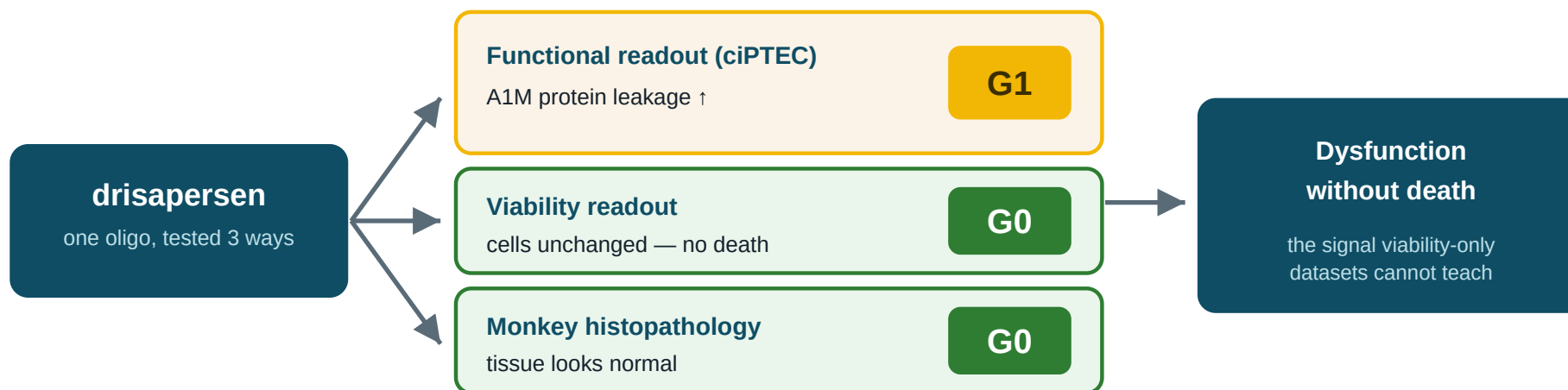
- **Wu 2022** · marketed-ASO nephrotoxicity review (*REV*)
- **inotersen** · NEJM NEURO-TTR + FDA label — severe anchor (*A1*)
- **mipomersen, volanesorsen** · gapmer renal signals (*A9, A8*)
- **inclisiran, givosiran, nusinersen** · safe controls
- **US 11,479,818 B2** · 2nd patent, queued (*N4*)

PART 4

# What the data shows

Three findings worth a biochemistry expert's attention

## Finding 1 — the “invisible” injury, captured



For the **same** drug we store **paired rows**: a mild grade on the functional test, a clean grade on viability/tissue. Side by side they encode “dysfunction without death” — a distinction viability-only datasets simply cannot teach a model.

## Finding 2 — a patent unlocked sequence + toxicity together

US 11,105,794 B2 — Table 1 (public domain)

| compound | sequence      | SEQ ID | in-vivo class |
|----------|---------------|--------|---------------|
| Cmpd 1-1 | CGTcag...AATC | 1      | low → G1      |
| Cmpd 1-2 | ACGtta...GGTA | 2      | medium → G2   |
| Cmpd 2-1 | TGCcat...TACG | 3      | high → G3     |
| ...      | ...           | ...    |               |

sequence AND toxicity in the SAME row — rare & valuable words mapped to our 0–3 scale

+21 measurement rows

our single biggest source  
in one clean, public-domain panel

sequences with real letters

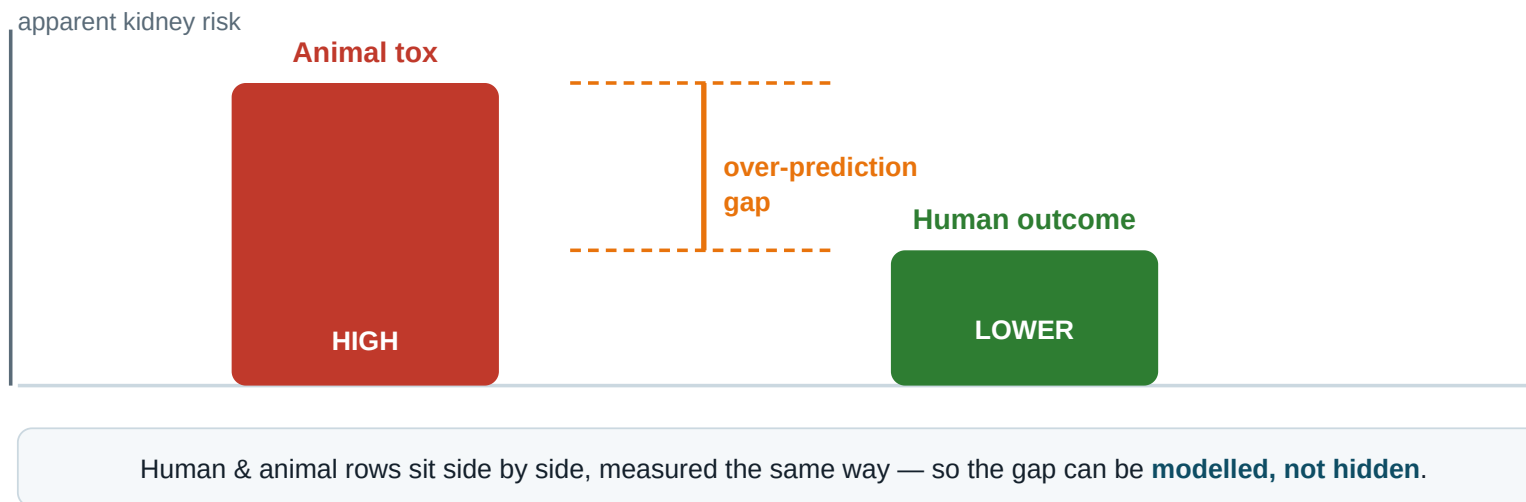
13 → 33

tripled — the most model-ready slice

Most sources give a sequence *or* a toxicity value. This public-domain patent table gave **both in the same row** for many oligos — the most directly model-ready slice of the dataset, and it tripled our sequence coverage.

## Finding 3 — animal tests over-predict human risk

Animal tests look worse than patients actually fare



A known, consistent bias for this drug class: animal kidneys look worse than patients turn out to be. We captured human and animal rows side by side so the gap can be **modelled, not ignored** — don't treat animal histopathology as human ground truth.

PART 5

# Why it fits Phase 2 — and what's next

Judging criteria, trust, limitations, and the ask

# Built for the four Phase-2 judging criteria

## 1 · Ability to solve

A **model-ready** design — granular **sequence + chemistry** predictors against **graded**, per-condition kidney outcomes; the patent panel even puts sequence **and** toxicity in one row.

## 2 · Potential impact

Fills a real public-domain gap: most open oligo-tox data is **hepatic** — kidney is thin. First open set linking oligo **design** to **graded nephrotoxicity**, incl. the functional phenotype.

## 3 · Feasibility & rigor

Strict **no-fabrication**, per-row provenance, controlled vocabularies and automated **QC** (0 orphans) — a set researchers can trust and reuse on release.

## 4 · Transparency & reproducibility

Open license (CC-BY) intended; every value **traceable** to source; all stats regenerate from data/ — **FAIR**, aligned to NIH data-sharing.

NCATS's four Phase-2 confidence factors — the dataset was **designed against them, not retro-fitted**. Submission window 1 May – 31 Dec 2026.



# Mapped to Phase-2's technical scoring rubric

## Experimental design

20 pts

**Controls built in** — 27 grade-0 **negative controls** (GalNAc-siRNA, intrathecal ASO, aptamer) and severe **positive anchors** (inotersen GN, SPC5001 AKI), plus within-compound functional-positive / structural-negative pairs. Source-study replicates stay traceable through per-row provenance.

## Data translatability

20 pts

**Built for modeling** — explicit **predictor variables** (sequence, chemistry, backbone, gapmer design, conjugate) against **graded toxicity indicators**; 7 modalities, full 0–3 range; captures the functional-not-cytotoxic signal and the animal → human gap. Patent rows carry sequence + grade in one record.

## Dataset management & documentation

10 pts

**FAIR by construction** — stable IDs, open CSV + CC-BY, controlled vocabularies (**common data elements**), a full data dictionary (`schema.md`), per-row provenance + redistribution, automated QC. A **Public Access & Dissemination Plan** (`PADP.md`) accompanies the submission.

The Evaluation Panel's technical criteria (≈50 pts shown) — complementing the four judging factors on the previous slide.

# Every row is defensible

## PROVENANCE — THE PAPER TRAIL

Each measurement carries its **source**, the **exact** table/figure/claim, and its **redistribution right**, so any value can be re-checked at its origin.

47 public-domain rows

64 summary-stat rows

16 sources

## QUALITY CONTROL — ALL PASSING

- ✓ every category value from an allowed list
- ✓ column counts intact (17 / 23)
- ✓ every result links to a real drug — **0 orphans**
- ✓ no duplicate IDs · grades all 0–3
- ✓ no-guessing sequence rule held

## Honest limitations

- **Grades are provisional** — assigned by rubric, awaiting expert sign-off.
- **Sequences 33/65** — remaining gaps (siRNA guide strands, some PMOs) left blank, never guessed.
- **36 “WS” rows** rest on web summaries — need verification against the primary document.
- **In-vitro human-cell rows (19)** are the scientific core but still a minority — growing them is the top priority.
- **Animal over-prediction** is present by design — it must be modelled, not ignored.

Stating the soft spots plainly is part of the deliverable — a reviewer should know exactly where to push.

# What we need from the biochemistry expert

- 1 **Grade sign-off.** Check the rubric → row mapping in `measurements.csv` so we can remove the “provisional” flag. Hot spots: the **grade 2 ↔ 3** line (biomarker rise vs. true AKI) and the **patent words** → **grade** mapping.
- 2 **Biology sanity check.** Is the functional-vs-structural story (megalin/cubilin → protein leak → grade 1) faithful, and are the readout → severity calls physiologically sound?
- 3 **Source confidence.** Flag any anchor — especially the 36 “WS” rows — you’d want re-verified against the primary text before release.

On hold until sign-off: the ≤12-page narrative · verifying WS rows · backfilling sequences · mining patent N4 for more rows.

## Where everything lives

|                       |                                                      |
|-----------------------|------------------------------------------------------|
| README.md             | strategy, scope, live record counter                 |
| schema.md             | data dictionary + the grade rubric                   |
| METHODOLOGY.md        | how it was built (sources → grading → QC)            |
| PADP.md               | public access & dissemination plan (licensing)       |
| PRESENTATION.md       | this deck (+ plain-English speaker notes)            |
| data/oligos.csv       | 65 drugs · 17 design columns                         |
| data/measurements.csv | <b>111 graded rows</b> ← <a href="#">review here</a> |
| sources/              | the source PDFs + the source registry                |

**Thank you — feedback welcome at the row level.** Every number in this deck regenerates from data/; nothing here is hand-maintained prose detached from the tables.